

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A factory makes **9**-inch, **7**-inch, and **4**-inch concrete screws. During a certain day, the number of **9**-inch concrete screws that the factory makes is **5** times the number *n* of **7**-inch concrete screws, and the number of **4**-inch concrete screws is **22**. During this day, the factory makes **100** concrete screws total. Which equation represents this situation?

Answer

- A. $9(5n) + 7n + 4(22) = 100$
- B. $9n + 7n + 4n = 100$
- C. $5n + 22 = 100$
- D. $6n + 22 = 100$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day at a factory, the number of **7**-inch concrete screws the factory makes is *n* and the number of **4**-inch concrete screws the factory makes is **22**. It's also given that during this day the number of **9**-inch concrete screws the factory makes is **5** times the number of **7**-inch concrete screws, or **5n**. Therefore, the total number of **7**-inch, **9**-inch, and **4**-inch concrete screws is $n + 5n + 22$, or $6n + 22$. It's given that during this day, the factory makes **100** concrete screws total. Thus, the equation $6n + 22 = 100$ represents this situation.

Choice A is incorrect. This equation represents a situation where the total length, in inches, of all the concrete screws, rather than the total number of concrete screws, is **100**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the total number of **9**-inch concrete screws and **4**-inch concrete screws, not including the **7**-inch concrete screws, is **100**.